Capstone Weekly Progress Report 4

ChalkStream (Group #62)

[Week of Nov 17, 2025]

Supervisor: Safaa Bedawi

Group Members: Cameron Bodkin-Dick, Eric Cui, Sandy Alzabadani

1. Overview of Current Week

Making progress towards integrating the frame pre-processing and content segmentation components, and compiling research and documentation of the existing modules. We aim to prepare information for the progress report, outlining the work we have completed, primarily in relation to the project proposal.

2. Milestone Progress Summary

Component	Description	Current Status % Complete	Remarks / Issues
3.1 Frame Preprocessing	Isolating the board from the rest of the wall, rectifying any angular distortion, and handling dynamic obstructions (instructor walking across the board)	70%	Problems with board isolation as our current scripts have trouble recognizing the board
3.2 Content Segmentation	Determine candidate areas of written content, and assign text, equation, and diagram affinity	85%	Running the PaddleOCR mobile model on a GPU has some unexpected dependencies.
3.3. Handwriting OCR	Setting up the testing & training environment for the OCR and prepping datasets.	50%	The environment for training the OCR is set up and ready to start.

3. Individual Member Contributions

Cameron Bodkin-Dick

This Week: Worked on organizing and acquiring documentation for developing the content segmentation and related modules. We are consolidating the information gathered through research for the Fall Progress Report, so that in future weeks, it will be easier to compile work products into the reports. Went back through the module to define any non-trivial functionalities and libraries. I wanted to have documentation detailing what each function and helper is responsible for and how it was done.

Next Week: Setting up the canvas edge detection for frame pre-processing and integrating with segmentation and an orchestration layer for live service demo.

Challenges: Going back through resources that I didn't save links to was a little painstaking to refind for sourcing; it’ll take a little more time.

Eric Cui

This Week:
Finalized Image Preprocessing Unit. Created a placeholder system for board resize that can resize frames with manual inputs. Integrated board resize, noise filter, and optionally grey scaling.
Additionally, it has been fully tested with various lecture and board colors. Lighting adjustments from the camera may disrupt the image, but it remains in reasonable range.

Next Week:
Begin in Diagram Vectorization

Challenges:
Board detection remains unstable and requires manual adjusting. However, future plans for improvements are prepared. (require further development of OCRs)

Sandy Alzabadani

This week:
I am in the progress of running my training for trOCR for handwritten data. I tried running the training for the database consisting of over 5000 images but that is infeasible to be done on CPU as it could take over 3 days and I would run out of memory. I took a sample of my database which is about 1000 images chosen at random from the

Next Week:
Research the processing requirements for training the TrOCR module and equation recognition model (later in the year) for request sent to Ryan Taylor from the sysc department.

Challenges: